

VEGU Complete Technical and Operations Documentation

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1. System Overview

- Customer App (web + Android wrapper)
- Rider App (web + Android wrapper)
- Admin Panel (owner controls)
- Operations Console (separate laptop/tablet page at /ops)
- Backend API (Node.js + Express + Prisma)
- Database (PostgreSQL via Prisma)

2. Technology Stack

- Frontend: Next.js 14, React, TypeScript, React Query, Tailwind CSS
- Android wrapper: Capacitor + Gradle
- Backend: Node.js, Express, TypeScript, Prisma ORM, Zod validation
- Database: PostgreSQL
- Auth: JWT access/refresh token architecture in backend auth controllers

3. Major Changes Completed

3.1 Android Studio test fix

- Updated package assertion in instrumented test to `com.vegu.app` .
- File:
`frontend/android/app/src/androidTest/java/com/getcapacitor/myapp/ExampleInstrumentedTest.java` .

3.2 Product + inventory model upgrade

Product model now includes:

- `brand`
- `minStockAlert`
- `purchasePrice`
- `batchNumber`
- `expiryDate`

File: `backend/prisma/schema.prisma` .

3.3 Admin product controls

- Add/Edit supports richer fields.
- Product hide/show endpoint added: `PATCH /api/admin/products/:id/availability` .
- Delete now performs hard delete with safe guard when referenced by orders/carts.
- Image update path supports URL or data image payload from admin product form.

3.4 Inventory controls

- Inventory API now returns cost, batch, expiry, low-stock threshold metadata.
- Stock update supports stock + purchase price + batch + expiry + alert threshold.

3.5 Separate operational store management app

- New dedicated operations route: `/ops` .
- Designed for laptop/tablet operations usage.
- Includes service pause/resume controls and quick links to all operational modules.

3.6 Timed service pause (15 minutes)

- Admin settings now supports timed pause via API.
- Endpoints:
- `POST /api/admin/settings/pause-service` with `{ minutes, reason }`
- `POST /api/admin/settings/resume-service`
- Order placement is blocked while maintenance/timed pause is active.

3.7 Expanded Indian grocery catalogue

- 14 categories, 45 realistic products.
- Two seeded stores/vendors: Nellore and Tirupati.
- Store-level assignment in seed data.

4. File-level Change Map (where to edit manually)

Backend

- Schema: `backend/prisma/schema.prisma`
- Seed data: `backend/prisma/seed.ts`
- Admin product APIs: `backend/src/controllers/admin/products.controller.ts`
- Admin inventory APIs: `backend/src/controllers/admin/inventory.controller.ts`
- Admin settings APIs: `backend/src/controllers/admin/settings.controller.ts`
- Order block during pause: `backend/src/controllers/order.controller.ts`
- Public config output: `backend/src/controllers/public.controller.ts`
- Admin routes: `backend/src/routes/admin.routes.ts`

Frontend

- Admin products UI: `frontend/src/app/admin/products/page.tsx`
- Admin inventory UI: `frontend/src/app/admin/inventory/page.tsx`
- Admin settings UI: `frontend/src/app/admin/settings/page.tsx`
- Ops app (separate): `frontend/src/app/ops/page.tsx`
- Public settings types: `frontend/src/lib/publicConfig.ts`

5. How to change key business settings manually

From UI (preferred)

- Open Admin: `/admin/settings`
- Update delivery fees, support info, maintenance mode, referral and wallet limits.
- Use pause button for 15-minute operational hold.
- Use Ops Console: `/ops` for quick operational controls on laptop/tablet.

From database

- Settings table stores key-value config at `settings` (Prisma model `Setting`).
- Keys include maintenance and timed pause values.

6. Build and run commands

Backend

- `cd vegu-app/backend`
- `npm install`
- `npm run db:push`
- `npm run db:seed`
- `npm run build`
- `npm run dev`

Frontend

- `cd vegu-app/frontend`
- `npm install`
- `npm run build`
- `npm run dev`

Android Studio

- Open `vegu-app/frontend/android` in Android Studio.
- Sync Gradle.
- Run test: `./gradlew testDebug`.

7. Validation checklist completed

- Backend build: pass
- Frontend build: pass
- Database push: pass
- Seed run: pass (14 categories, 45 products)
- Android test (previously failing): pass

8. Production notes

- Current upload path supports URL/data-image payloads; production media pipeline (Cloudinary/S3/Firebase Storage + CDN transforms) is recommended next.
- Timed pause currently blocks new order placement from backend order API for operational safety.
- Ops console is admin protected and intended for real store operations usage on larger screens.